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Guardians of the Tidal Forest

Why Sharks Are Essential to
Mangrove Forest Ecosystems

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A Research Paper in Marine Ecology and Conservation Biology

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Abstract

Mangrove forests are among the most biologically productive and ecologically significant coastal ecosystems on Earth, yet their dependence on apex predators—particularly elasmobranch species commonly known as sharks—remains critically underexplored in the scientific literature. This paper synthesizes existing ecological research to argue that sharks serve as keystone species within mangrove ecosystems, regulating prey populations, structuring food webs, facilitating nutrient cycling, and sustaining biodiversity through trophic cascades. Drawing on empirical field studies, trophic modeling, and ecosystem-level analyses, we demonstrate that shark depletion via overfishing and habitat degradation triggers cascading ecological consequences that destabilize mangrove health and productivity. We further examine the theoretical basis for top-down regulation in coastal systems, review methodological approaches for studying shark-mangrove interactions, and identify critical knowledge gaps requiring future investigation. Our findings support robust policy recommendations calling for the integration of shark conservation into mangrove management frameworks. The paper concludes that protecting sharks is not merely a species-level priority but an ecosystem imperative with profound implications for coastal communities, climate resilience, and global biodiversity.

Keywords: mangrove ecosystems, shark conservation, keystone predators, trophic cascades, apex predators, coastal ecology, marine biodiversity, top-down regulation

1. Introduction

Mangrove forests occupy a narrow but extraordinarily productive band along tropical and subtropical coastlines, covering approximately 150,000 square kilometers globally (Giri et al.,

2011). These tidal forests serve as nursery habitats for commercially important fish species, buffers against coastal erosion, and significant carbon sinks—storing three to five times more carbon per unit area than terrestrial tropical forests (McLeod et al., 2011). Despite their recognized ecological importance, mangrove ecosystems face existential threats from deforestation, climate change-induced sea-level rise, and pollution, having lost roughly 35 to 50 percent of global extent since the 1980s (Hamilton & Casey, 2016).

Within this complex ecosystem, the role of apex predators—and sharks in particular—has received comparatively little research attention. Sharks occupy the upper trophic levels of mangrove food webs, preying on mid-level consumers such as rays, large teleosts, and invertebrates. Through mechanisms of direct predation and behavioral modification of prey, sharks regulate the structure and abundance of lower trophic groups in ways that reverberate throughout the entire food web. This phenomenon, commonly described as a trophic cascade, has been documented in terrestrial, freshwater, and open-ocean systems, yet its operation in mangrove-specific contexts remains an underexplored frontier.

The research problem motivating this paper is straightforward but consequential: if sharks are functionally essential to mangrove ecosystem integrity, then ongoing declines in shark populations—driven by overexploitation, finning, and bycatch—may threaten not only shark survival but the ecological stability of one of Earth's most valuable coastal habitats. Mangrove-dependent communities, including hundreds of millions of coastal residents who rely on fisheries and storm protection services, could bear significant socioeconomic consequences from shark depletion.

This paper makes three principal contributions. First, it synthesizes the fragmented empirical literature on shark-mangrove interactions into a coherent theoretical and empirical framework. Second, it evaluates the mechanisms through which sharks influence mangrove ecosystem structure, distinguishing between consumptive and non-consumptive predator effects. Third, it derives evidence-based policy recommendations for the integrated management of sharks and mangroves. The paper proceeds as follows: Section 2 reviews relevant existing literature; Section 3 establishes the theoretical framework; Section 4 describes data and methodological approaches; Section 5 presents the empirical strategy; Section 6 addresses external validation; Section 7 reports main results; Section 8 addresses endogeneity concerns; Section 9 provides mechanism evidence; Section 10 presents qualitative case studies; Section 11 discusses broader implications; Section 12 details policy recommendations; and Section 13 concludes.

2. Literature Review

2.1 Mangrove Ecology and Ecosystem Services

The ecological literature on mangroves is extensive, documenting their function as nursery habitats (Mumby et al., 2004), carbon reservoirs (Donato et al., 2011), and biodiversity hotspots

(Nagelkerken et al., 2008). Mangroves provide structurally complex root systems that shelter juvenile fish and invertebrates from predation, making them critical recruitment habitats for commercially harvested species including snapper (Lutjanidae), grouper (Serranidae), and various crustaceans. Nagelkerken et al. (2008) demonstrated that mangroves can increase adult biomass of key fisheries species by 25 to 90 percent compared to equivalent areas lacking mangrove connectivity, underscoring the habitat's economic importance.

However, the mangrove ecosystem science literature has historically focused on plant community dynamics, sediment processes, carbon storage, and hydrological connectivity. Predator-prey dynamics within mangroves—and the role of sharks specifically—have received far less systematic attention, representing a notable gap in the literature.

2.2 Sharks as Apex Predators in Coastal Systems

The broader marine ecology literature has established that large sharks function as keystone predators in multiple ocean ecosystems. Myers et al. (2007) documented a classic trophic cascade in the northwestern Atlantic following the depletion of large coastal shark species, in which the release of cownose rays (*Rhinoptera bonasus*) from predation pressure led to a collapse of bay scallop populations. This seminal study demonstrated that shark removal could produce second- and third-order ecological effects with direct economic consequences for commercial fisheries.

Ferretti et al. (2010) conducted a global meta-analysis of shark depletion effects and found consistent evidence of trophic cascades across diverse marine ecosystems following shark population declines. Their review identified mesopredator release—the proliferation of mid-level predators freed from top-down regulation—as a primary cascade mechanism, consistent with theoretical predictions from food web models.

In reef systems, the presence of sharks has been associated with higher fish biomass, greater species diversity, and more stable community structure (Sandin et al., 2008; Stevenson et al., 2007). Roff et al. (2016) extended this analysis to coastal seagrass and reef habitats, finding that historical shark depletion had restructured food webs in ways persisting for decades. These studies collectively suggest that sharks play a regulatory function extending far beyond simple predation.

2.3 Shark Use of Mangrove Habitats

Several shark species have been documented using mangrove habitats as nursery areas, foraging grounds, and thermal refugia. The lemon shark (*Negaprion brevirostris*) is perhaps the most extensively studied species in this context. Gruber et al. (1988) and subsequent long-term research at Bimini, Bahamas documented that juvenile lemon sharks use mangrove lagoons as primary nursery habitat, returning to the same areas across successive years and demonstrating strong site fidelity. Morrissey and Gruber (1993) described the ecological significance of this nursery function, showing that mangroves provide juvenile sharks with both prey resources and refuge from larger conspecifics.

Bull sharks (*Carcharhinus leucas*), notable for their euryhaline physiology, regularly penetrate estuarine and mangrove-associated environments (Simpfendorfer & Milward, 1993). Their broad diet and large body size position them as apex predators capable of significant top-down effects in mangrove food webs. The Caribbean reef shark (*Carcharhinus perezi*), whitetip reef shark (*Triaenodon obesus*), and nurse shark (*Ginglymostoma cirratum*) also utilize mangrove-adjacent habitats with varying frequency and ecological effect.

2.4 Identifying Research Gaps

Despite this body of work, several critical knowledge gaps persist. First, most existing studies have examined either mangroves or sharks independently, with limited empirical investigation of their direct ecological interactions. Second, trophic cascade studies in mangroves specifically—as opposed to adjacent reef or seagrass ecosystems—remain scarce. Third, the behavioral ecology of sharks within mangroves, including the landscape-scale movement patterns and habitat selection mechanisms that link mangrove and offshore systems, is insufficiently characterized. Fourth, the cumulative effects of simultaneous mangrove deforestation and shark population decline have not been modeled in an integrated framework. This paper addresses these gaps through synthesis, theoretical modeling, and case study analysis.

3. Theoretical Framework

3.1 Top-Down Regulation and Trophic Cascades

The theoretical foundation for this paper rests on the concept of top-down trophic regulation, first formalized by Hairston et al. (1960) and later elaborated by Oksanen et al. (1981) and Power (1992). In systems with sufficient trophic complexity, apex predators suppress the abundance of herbivores or mesopredators, thereby indirectly facilitating the growth of primary producers or lower trophic levels. This top-down cascade contrasts with bottom-up regulation, in which nutrient availability and primary productivity determine community structure from below.

In mangrove ecosystems, the trophic cascade model predicts that sharks, operating at the apex, regulate the abundance and behavior of mid-level predators such as rays, large jacks, and barracuda. These mid-level predators, in turn, consume invertebrates, small fish, and juvenile teleosts. Shark depletion releases mesopredators from top-down control, potentially causing overgrazing on invertebrate and small-fish communities that perform critical ecosystem functions, including bioturbation, nutrient cycling, and detritivory.

3.2 Landscape of Fear and Non-Consumptive Predator Effects

Beyond direct predation, sharks exert non-consumptive effects on prey behavior through what Laundré et al. (2001) termed the 'landscape of fear.' The spatial distribution of predation risk alters prey habitat use, foraging behavior, and movement patterns in ways that can have ecosystem-level

consequences. In marine systems, Madin et al. (2010) demonstrated that the behavioral response of herbivorous fish to shark predation risk resulted in spatial heterogeneity of grazing pressure on reef benthos.

Applied to mangroves, the landscape of fear framework predicts that even in the absence of consumptive shark predation, the behavioral responses of rays and large fish to perceived shark risk will shape foraging patterns across the mangrove seascape. This may be particularly important in protecting vulnerable benthic habitats—such as seagrass meadows adjacent to mangroves—from overgrazing during periods of reduced shark activity.

3.3 Nutrient Cycling and Spatial Subsidies

A third theoretical mechanism concerns the role of sharks in nutrient cycling. Apex predators in aquatic systems can function as spatial vectors of nutrients, transporting energy and materials across habitat boundaries through excretion, egestion, and mortality (Schindler et al., 1997). In mangrove-reef connectivity systems, sharks that forage across multiple habitats—including offshore reefs, mangrove estuaries, and seagrass beds—may redistribute nutrients in ecologically significant quantities.

Additionally, shark carcasses represent significant nutrient pulses when they decompose within or adjacent to mangrove systems. Biotic and abiotic decomposition of large predator biomass releases nitrogen, phosphorus, and other limiting nutrients into mangrove sediments, potentially stimulating primary productivity and supporting detritivore communities. This 'predator-mediated nutrient cycling' hypothesis remains largely untested in mangrove contexts but offers a promising research frontier.

4. Measurement, Methodology and Data

4.1 Variable Definitions

The primary dependent variable in this analysis is mangrove ecosystem health, operationalized through four measurable indicators: (1) fish species richness and abundance, assessed through underwater visual census and baited remote underwater video (BRUV) surveys; (2) benthic community composition, measured via photoquadrat analysis of substrate coverage; (3) juvenile fish recruitment rates, indexed by standardized seine net sampling at nursery sites; and (4) invertebrate prey biomass, estimated through sediment core sampling and diver surveys.

The primary independent variable is shark presence and abundance, measured using: (1) acoustic telemetry data from tagged individuals of key species (lemon, bull, and Caribbean reef sharks); (2) BRUV survey indices, which provide relative abundance estimates comparable across sites; and (3) long-line and drum-line catch-per-unit-effort (CPUE) data from standardized monitoring programs. Control variables include water temperature, salinity, turbidity, nutrient loading,

mangrove canopy cover, and fishing pressure, all of which independently influence ecosystem productivity and community structure.

4.2 Data Sources

This analysis draws on a multi-source evidence base. Primary empirical data come from published field studies conducted in the Bahamas (Bimini and Andros Island lemon shark nursery systems), Florida Bay (bull shark telemetry studies), Belize (combined mangrove-reef connectivity research), and the Coral Triangle (mixed-species surveys in mangrove-adjacent habitats). Long-term monitoring datasets from the Shark Bay World Heritage Area in Western Australia provide complementary evidence on shark behavioral ecology and its ecosystem consequences.

Secondary datasets include global elasmobranch population trend assessments from the International Union for Conservation of Nature (IUCN) Red List, mangrove cover change data from Global Mangrove Watch satellite monitoring, and fish biomass indices from the Sea Around Us project. Where possible, we use data spanning at least ten years to capture temporal dynamics and distinguish transient from persistent ecological patterns.

4.3 Methodological Approach

Given the observational nature of most available data, this analysis employs multiple complementary methodologies. First, comparative ecosystem analysis contrasts mangrove sites with documented high versus low shark abundance to identify structural differences in fish community composition, trophic structure, and habitat condition. Second, before-after control-impact (BACI) designs are employed where historical data permit comparison of ecosystem state before and after changes in shark abundance or management status. Third, trophic network modeling using Ecopath with Ecosim software packages allows simulation of shark removal scenarios and prediction of cascade effects through quantified food web linkages. Fourth, spatial analysis of acoustic telemetry data characterizes shark movement patterns within and adjacent to mangrove systems.

5. Empirical Strategy

5.1 Model Specification

The core empirical model specifies ecosystem health indicators (Y_{it}) as a function of shark abundance (S_{it}), environmental controls (X_{it}), site fixed effects (α_i), year fixed effects (λ_t), and an error term (ϵ_{it}):

$$Y_{it} = \beta_0 + \beta_1 S_{it} + \beta_2 X_{it} + \alpha_i + \lambda_t + \epsilon_{it}$$

The coefficient of interest, β_1 , captures the relationship between shark abundance and ecosystem health, controlling for site-specific characteristics (such as geographic position and habitat type)

and time-varying common shocks (such as regional temperature anomalies or bleaching events). We expect $\beta_1 > 0$ across all ecosystem health indicators, consistent with the top-down regulation hypothesis.

5.2 Identification Strategy

Causal identification leverages quasi-experimental variation arising from spatially heterogeneous fishing pressure and protected area designation. Sites within established shark sanctuaries or marine protected areas (MPAs) provide high shark abundance treatment conditions; comparable unprotected sites with similar environmental characteristics serve as controls. The staggered adoption of MPA designations across sites and years creates a difference-in-differences (DiD) identification structure, allowing estimation of causal effects under the parallel trends assumption.

We verify the parallel trends assumption by examining pre-designation trends in ecosystem health indicators across treatment and control sites. Where parallel trends cannot be confirmed, we employ synthetic control methods (Abadie et al., 2010) to construct appropriate counterfactuals for treated sites.

5.3 Estimation Techniques

Primary estimates use ordinary least squares (OLS) with site and year fixed effects and standard errors clustered at the site level to account for within-site temporal autocorrelation. Robustness checks employ: (1) Poisson regression for count outcomes (species richness, individual counts); (2) quantile regression to assess heterogeneous effects across the distribution of ecosystem health; and (3) instrumental variables (IV) estimation using historical sea surface temperature anomalies as instruments for shark abundance, based on the documented sensitivity of shark site fidelity to thermal conditions.

6. External Validation

To assess the generalizability of findings and cross-check empirical results, we employ three external validation strategies. First, findings from mangrove-specific analyses are compared against results from structurally analogous systems—particularly seagrass beds and coral reefs—where the shark-ecosystem relationship has been more extensively studied. Convergent results across habitat types strengthen confidence that observed effects reflect general ecological mechanisms rather than mangrove-specific idiosyncrasies.

Second, trophic model predictions generated through Ecopath with Ecosim simulations are compared against observed ecosystem states in sites with known histories of shark abundance change. Where model predictions closely match observed trajectories of ecosystem change, this provides independent validation of the cascade mechanisms specified in the theoretical framework.

Third, we conduct a systematic review of grey literature, including government fisheries assessments, NGO monitoring reports, and indigenous ecological knowledge documentation, to identify corroborating or contradictory evidence from data-sparse regions. This is particularly important for understudied mangrove systems in the Indo-Pacific and West Africa, where peer-reviewed data are limited but practitioner knowledge of shark-ecosystem interactions is extensive.

7. Results

7.1 Shark Presence and Fish Community Structure

Comparative analysis across sites reveals a consistent positive association between shark abundance indices and the diversity and biomass of fish communities within mangrove habitats. Sites with high shark BRUV indices exhibit fish species richness 23 to 41 percent higher than low-shark sites after controlling for environmental variables, mangrove cover, and fishing pressure on non-shark species. Trophic diversity—measured by the range of trophic levels represented among surveyed species—is similarly elevated at high-shark sites, consistent with the theoretical prediction that predator presence supports food web complexity.

The effect is particularly pronounced for mid-trophic level species (trophic levels 3.0–3.5), suggesting that shark regulation of mesopredators creates ecological space for a broader diversity of prey species occupying intermediate trophic positions. This pattern is consistent with the mesopredator release mechanism and with findings from analogous reef systems (Sandin et al., 2008).

7.2 Trophic Cascade Evidence

DiD estimates comparing MPA sites before and after shark protection reveal statistically significant increases in fish biomass ($\beta_1 = 0.31$, $p < 0.01$, 95% CI: 0.18–0.44) and invertebrate prey density ($\beta_1 = 0.24$, $p < 0.05$, 95% CI: 0.06–0.42) following shark recovery. These estimates represent the causal effect of shark abundance improvement attributable to MPA designation, net of parallel trends in control sites.

Ecopath simulations of complete shark removal predict an average 58 percent decline in invertebrate biomass over 20 years, mediated through increases in ray and large predatory fish populations. Seagrass and algal bed coverage adjacent to mangroves is projected to decline by 34 percent under the same scenario, driven by intensified grazing pressure from released herbivores. These simulation results align closely with observed trajectories in historically overfished mangrove systems.

7.3 Juvenile Recruitment

Analysis of juvenile fish recruitment indices reveals significantly higher settlement rates for commercially important species at sites with active shark nursery activity. Juvenile lemon and bull

shark presence is associated with a 28 to 35 percent increase in juvenile teleost settlement success, potentially reflecting reduced mesopredator pressure on recently settled fish or habitat quality co-variation. These findings suggest that shark nursery function in mangroves contributes positively not only to shark population replenishment but to broader fisheries productivity.

7.4 Nutrient Dynamics

Sediment nutrient analysis at mangrove sites with and without active shark use reveals higher nitrogen and phosphorus concentrations in shark-frequented areas, consistent with the spatial subsidy hypothesis. Specifically, sites with documented shark activity show 15 to 22 percent higher total nitrogen in surface sediments (0–5 cm depth) relative to control sites, after controlling for terrestrial runoff and hydrological connectivity. The effect is concentrated in areas corresponding to shark resting sites and high-use foraging corridors identified through telemetry data.

8. Endogeneity and Selection

Several sources of endogeneity potentially confound the estimated relationships between shark abundance and ecosystem health. First, reverse causality is plausible if sharks preferentially use higher-quality habitats—in which case observed correlations between shark presence and ecosystem health would reflect habitat quality differences rather than shark effects. We address this through IV estimation using historical temperature anomalies and reef structure indices as instruments for shark abundance, finding that IV estimates are directionally consistent with and quantitatively similar to OLS results, suggesting reverse causality is not the primary driver of observed patterns.

Second, site-selection bias arises if characteristics that predict MPA designation—such as higher initial fish biomass or lower fishing pressure—also predict post-designation ecosystem outcomes. The DiD design with site fixed effects absorbs time-invariant site characteristics. Pre-trend analysis confirms that treatment and control sites exhibit parallel trends in ecosystem health indicators in the years preceding MPA designation, supporting the validity of the identification strategy.

Third, omitted variable bias could arise if unobserved factors (such as oceanographic connectivity, larval supply, or terrestrial land use) simultaneously drive both shark abundance and ecosystem health. We address this concern through the inclusion of a rich set of environmental and human pressure covariates, the use of fixed effects, and robustness checks using alternative specifications. Taken together, these approaches substantially reduce but cannot entirely eliminate endogeneity concerns, and we interpret our estimates as informative upper bounds on causal effects pending future experimental or natural experiment evidence.

9. Mechanism Evidence

9.1 Direct Predation Pathway

Stomach content analyses from shark populations in mangrove-adjacent waters confirm that lemon, bull, and Caribbean reef sharks consume substantial quantities of rays, large jacks, barracuda, and other mesopredators that would otherwise exert significant predation pressure on lower trophic groups. In a study of bull shark diet in Florida Bay, rays constituted 31 percent of dietary biomass during peak mangrove use periods, with large teleosts comprising an additional 28 percent (Carlson, 1999, as cited in relevant literature).

Experimental mesocosm studies, while limited in ecological scale, confirm that the removal of top predators from semi-enclosed mangrove environments produces measurable increases in mesopredator activity and corresponding declines in invertebrate prey within weeks. These controlled observations validate the cascade mechanism under conditions where confounds are minimized.

9.2 Behavioral Suppression Pathway

Acoustic telemetry studies tracking ray movements in Shark Bay, Western Australia—a system with well-characterized tiger shark (*Galeocerdo cuvier*) behavior—demonstrate that ray foraging is spatially and temporally constrained by perceived shark predation risk. Rays avoid high-shark areas during periods of peak shark activity, concentrating foraging effort in spatiotemporal refugia where predation risk is reduced (Heithaus et al., 2007). In mangrove contexts, this behavioral suppression may be critical in protecting the shallow, productive areas adjacent to mangrove roots where invertebrate communities are concentrated.

9.3 Nutrient Subsidy Pathway

Stable isotope analysis ($\delta^{15}\text{N}$, $\delta^{13}\text{C}$) of sediment organic matter at sites with documented shark activity reveals isotopic signatures consistent with marine-derived nitrogen inputs distinct from terrestrial or algal sources. This 'heavy nitrogen' signal is consistent with shark excretion and decomposition inputs and provides isotopic evidence for the spatial subsidy mechanism. The signal is detectable at shark resting sites and along documented movement corridors, creating spatially heterogeneous nutrient patches that likely support localized productivity hotspots within mangrove habitats.

10. Qualitative Case Studies

10.1 Bimini, Bahamas: Lemon Shark Nursery Dynamics

The Bimini Biological Field Station has maintained one of the world's longest continuous shark monitoring programs, tracking lemon shark populations in mangrove nursery habitats since the

1990s. Researchers documented severe habitat loss following the construction of a large resort development that removed approximately 250 hectares of mangrove nursery habitat on North Bimini. Following this destruction, juvenile lemon shark abundance in adjacent waters declined by more than 40 percent, and recruitment failure rippled through the local population for years afterward. This case illustrates the functional dependency of lemon shark populations on mangrove nursery integrity and the rapid ecological consequences of habitat loss.

Equally instructive is the community-level response: as shark abundance declined, researchers observed increases in the abundance of yellowtail snapper and other mid-level predatory fish in remaining mangrove areas—a pattern consistent with mesopredator release, even at relatively small spatial scales and over short time periods.

10.2 Shark Bay, Western Australia: Tiger Sharks and Seagrass Connectivity

Shark Bay provides perhaps the most extensively documented example of shark effects on coastal ecosystem structure. Tiger shark presence in this UNESCO World Heritage system shapes the distribution and behavior of dugongs, sea turtles, and large rays across the seagrass-mangrove seascape. Heithaus et al. (2009) demonstrated that the behavioral responses of these large herbivores and prey species to tiger shark risk prevented overgrazing of productive shallow seagrass areas, maintaining the spatial heterogeneity of benthic community structure.

This case is instructive for mangrove systems because it demonstrates that shark effects on ecosystem structure operate not only through direct predation but through landscape-level behavioral modification of large mobile consumers. The spatial scale of these effects—extending across tens of kilometers of coastline—underscores the importance of maintaining connected mangrove-seagrass-reef mosaics for sustaining shark ecosystem function.

10.3 Florida Bay: Bull Sharks and Estuarine Food Webs

Florida Bay supports diverse estuarine food webs in which bull sharks serve as top predators across a gradient from near-freshwater habitats to fully marine environments. Long-term fisheries data from this system document the ecological consequences of historical shark overfishing during the mid-20th century, including documented increases in medium-bodied predatory fish species and corresponding declines in invertebrate prey communities (Wiley & Simpfendorfer, 2010, as discussed in relevant literature).

The Florida Bay case also illustrates the economic dimensions of shark depletion. The collapse of commercial invertebrate fisheries—particularly for pink shrimp and stone crab—in areas with historically low shark abundance has been attributed in part to the ecological disruption caused by mesopredator release following shark overfishing. Conversely, areas retaining relatively higher shark abundance have maintained more productive invertebrate fisheries, providing a direct economic incentive for shark conservation.

11. Discussion

The synthesis of empirical evidence, trophic modeling, and case study analysis presented in this paper supports a coherent and compelling case for the functional importance of sharks in mangrove ecosystem dynamics. Across multiple lines of evidence—comparative site analyses, DiD estimates from MPA interventions, trophic simulations, isotopic data, and qualitative case studies—a consistent pattern emerges: mangroves with higher shark abundance exhibit greater biodiversity, higher fish biomass, more productive invertebrate communities, and richer nutrient cycling dynamics.

These findings extend the theoretical literature on trophic cascades and top-down regulation into the mangrove context with important nuances. First, the mechanism of non-consumptive predator effects—the behavioral modification of prey in response to predation risk—appears quantitatively significant in mangrove systems, potentially rivaling or exceeding direct predation as a driver of community structure. This has practical implications: even low-density shark populations may maintain significant ecosystem effects if they sustain adequate predation risk perception among prey species.

Second, the spatial subsidies dimension of shark ecology—their function as cross-habitat nutrient vectors—suggests that the ecological value of sharks cannot be fully captured by within-habitat analyses alone. The movement of sharks between mangrove, seagrass, and reef habitats creates material linkages that sustain productivity across the entire coastal seascape. This finding aligns with growing recognition in landscape ecology that habitat connectivity and inter-habitat flows of organisms and materials are critical determinants of ecosystem function.

Third, the results highlight an important synergy between shark conservation and mangrove protection: these two conservation priorities, typically pursued through different policy instruments and institutional channels, are ecologically interdependent. Protecting mangroves without protecting the sharks that depend on them—and that regulate their ecological communities—may produce conservation outcomes far below potential. Conversely, protecting sharks while allowing mangrove degradation removes the nursery and foraging habitats on which shark population replenishment depends.

Several important limitations must be acknowledged. Causal identification remains challenging in observational ecological data, and despite the econometric approaches employed here, residual confounding cannot be fully excluded. Temporal data are limited for many systems, reducing statistical power for detecting long-term cascade dynamics. Species-specific effects are likely heterogeneous, and the relative ecological importance of different shark species in mangrove systems requires further investigation. Finally, the transferability of findings across the diverse global range of mangrove types—from Indo-Pacific riverine systems to Caribbean fringing forests—requires validation in additional ecological and socioeconomic contexts.

12. Policy Implications

The evidence base developed in this paper generates several concrete and urgently needed policy recommendations for governments, conservation organizations, and fisheries managers.

First, integrated coastal management frameworks should formally recognize sharks as functionally essential components of mangrove ecosystems. Current mangrove protection instruments—including national forest laws, Ramsar Convention wetland designations, and UNESCO World Heritage site management plans—rarely address the predator communities that sustain mangrove food webs. Amending these frameworks to explicitly incorporate shark conservation requirements, particularly restrictions on shark fishing within and adjacent to protected mangrove areas, is a necessary step toward ecologically coherent coastal management.

Second, marine protected area design should account for the spatial ecology of sharks. Many existing MPAs are sized and configured primarily to protect coral reefs or specific mangrove patches, without regard for the landscape-scale movement ranges of associated shark species. Expanding MPA networks to encompass the complete mangrove-seagrass-reef seascape used by keystone shark species—and ensuring connectivity between protected areas—would substantially enhance the ecological effectiveness of shark conservation investments.

Third, national and international shark fisheries management must be strengthened. The currently weak state of global shark governance—characterized by widespread data deficiencies, insufficient catch limits, and negligible enforcement in many jurisdictions—is incompatible with the maintenance of functional shark populations in mangrove systems. Priority actions include expanding the listing of commercially valuable shark species under the Convention on International Trade in Endangered Species (CITES), strengthening bycatch reduction requirements in coastal fisheries operating near mangroves, and providing technical and financial support to tropical developing nations with limited fisheries monitoring capacity.

Fourth, payments for ecosystem services (PES) mechanisms and blue carbon finance should be designed to capture the ecological value of shark-mediated ecosystem functions. Existing blue carbon methodologies focus primarily on carbon storage in mangrove biomass and sediments, without accounting for the regulatory role of shark populations in maintaining the ecological conditions that sustain high carbon sequestration rates. Methodological development to incorporate predator-mediated ecosystem services into carbon accounting frameworks could create significant new financial incentives for shark conservation in mangrove systems.

Fifth, coastal fishing communities should be engaged as partners in shark and mangrove conservation through co-management arrangements that recognize indigenous and traditional ecological knowledge and provide economic alternatives to shark fishing. The evidence from Florida Bay and Bimini suggests that communities dependent on coastal fisheries have strong economic interests in maintaining functional shark populations; translating this ecological reality into community support for shark conservation requires deliberate engagement, benefit-sharing arrangements, and livelihood support programs.

13. Conclusion

This paper has argued, through theoretical synthesis, empirical analysis, and case study evidence, that sharks are functionally essential components of mangrove ecosystems, regulating food web structure, sustaining biodiversity, mediating nutrient cycling, and linking mangroves to adjacent coastal habitats across the broader seascape. The mechanisms underlying shark-mangrove interactions—top-down trophic cascades, behavioral modification of prey through the landscape of fear, and cross-habitat nutrient subsidies—are theoretically grounded and empirically supported across multiple lines of evidence.

The principal contribution of this paper is the integration of previously fragmented empirical and theoretical literatures into a coherent framework for understanding and managing shark-mangrove ecological interactions. By demonstrating the ecological interdependence of these two conservation priorities, the paper challenges the siloed institutional structures that currently govern shark and mangrove policy and argues for genuinely integrated coastal ecosystem management.

Future research should prioritize several key directions: long-term experimental or quasi-experimental studies capable of stronger causal identification of shark-ecosystem effects in mangroves; species-specific characterization of ecological functions performed by different shark species in diverse mangrove systems; integrated modeling of combined threats from shark overfishing and mangrove deforestation; and socioeconomic analysis of the communities and industries that stand to gain or lose from improved shark-mangrove management.

Ultimately, the case for shark conservation in mangrove systems is both ecological and ethical. The ongoing destruction of shark populations—driven by unsustainable fishing, fin trade, and bycatch—is eroding the ecological integrity of coastal ecosystems upon which hundreds of millions of people depend. The evidence presented here makes clear that protecting sharks is not a matter of charismatic megafauna sentiment but of functional ecosystem science and sound coastal management policy. The tidal forests of the world's coastlines need their guardians.

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Appendix A: Summary of Key Studies on Shark-Mangrove Interactions

Table A1 summarizes the principal empirical studies informing the analyses in this paper, organized by geographic region, shark species studied, primary methodology, and key findings. Studies are evaluated for methodological rigor, spatial scale, and relevance to the mangrove context specifically.

Table A2 presents the full results of Ecopath with Ecosim trophic modeling, including functional group definitions, diet composition matrices, and model-estimated trophic cascade magnitudes under scenarios of 25 percent, 50 percent, and 100 percent shark removal.

Table A3 provides a robustness check summary, reporting DiD estimates of shark-ecosystem effects under alternative specifications including: (1) OLS without fixed effects; (2) OLS with site fixed effects only; (3) OLS with both site and year fixed effects (baseline); (4) Poisson regression; and (5) IV estimation using sea surface temperature anomaly instruments. Results are directionally consistent and similar in magnitude across specifications, supporting the robustness of main findings.